In [1]:
```
from array import *
arr = array('i',[])

n = int(input('Enter the length of the array'))

for i in range(5):
    x = int(input('Enter the next value'))
    arr.append(x)
print(arr)
```

```
array('i', [7, 9, 8, 7, 6])
```

In [2]:
```
from array import *
arr = array('i',[])

n = int(input('Enter the length of the array'))

for i in range(5):
    x = int(input('Enter the next value'))
    arr.append(x)
print(arr)
```

```
array('i', [6, 7, 8, 8, 0])
```

In [3]:
```
from array import *
arr = array('i',[])

n = input('Enter the length of the array')

for i in range(5):
    x = input('Enter the next value')
    arr.append(x)
print(arr)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[3], line 8
      6 for i in range(5):
      7     x = input('Enter the next value')
----> 8     arr.append(x)
      9 print(arr)

TypeError: 'str' object cannot be interpreted as an integer
```

In [4]:
```
# Way of creating array using numpy
```

In [5]:
```
from numpy import *
arr = array([1,2,3,4,5])
print(arr)
type(arr)
```

```
[1 2 3 4 5]
```

Out[5]:
```
numpy.ndarray
```

In [6]:
```
print(arr.dtype)
```

```
int32
```

In [7]:
```
arr = array([1,2,3,4,5.9])
print(arr)
```

```
[1.  2.  3.  4.  5.9]
```

In [8]:
```
print(arr.dtype)
```

```
float64
```

In [9]:
```
arr2 = array([1,2,3,4,5.9],float)
arr2
```

Out[9]:
```
array([1. , 2. , 3. , 4. , 5.9])
```

In [10]:
```
arr3 = array([1,2,3,4,5.6],int)
arr3
```

Out[10]:
```
array([1, 2, 3, 4, 5])
```

In [11]:
```
import numpy as np
```

In [12]:
```
arr4 = np.linspace(0, 16, 10) # break the code between 10 spaces between 0 to 16
arr4
```

Out[12]:
```
array([ 0.        ,  1.77777778,  3.55555556,  5.33333333,  7.11111111,
        8.88888889, 10.66666667, 12.44444444, 14.22222222, 16.        ])
```

In [13]:
```
arr5 = np.arange(0,10,2) # arange - as range
arr5
```

Out[13]:
```
array([0, 2, 4, 6, 8])
```

In [14]:
```
arr6 = np.zeros(5)
arr6
```

Out[14]:
```
array([0., 0., 0., 0., 0.])
```

In [15]:
```
arr7 = np.ones(5)
arr7
```

Out[15]:
```
array([1., 1., 1., 1., 1.])
```

In [ ]:
```
import tkinter as tk
# Function to be called when the button is clicked
def on_button_click():
    label.config(text="Button clicked!")

# Create the main application window
root = tk.Tk()
root.title("Simple Tkinter App")

# Create a label widget
label = tk.Label(root, text="Hello, Tkinter!")
label.pack(pady=20)

# Create a button widget
button = tk.Button(root, text="Click Me", command=on_button_click)
button.pack(pady=20)
```

```
# Run the application
root.mainloop()
```

In [ ]:
```
import tkinter as tk
from tkinter import messagebox

# Function to be called when the button is clicked
def on_button_click():
    user_input = entry.get()
    messagebox.showinfo("Information", f"You entered: {user_input}")

# Create the main application window
root = tk.Tk()
root.title("Simple Tkinter App")

# Create a label widget
label = tk.Label(root, text="Enter something:")
label.pack(pady=10)

# Create a text entry widget
entry = tk.Entry(root, width=30)
entry.pack(pady=10)

# Create a button widget
button = tk.Button(root, text="Submit", command=on_button_click)
button.pack(pady=10)

# Run the appication
root.mainloop()
```

In [ ]:
```
__name__
```

In [ ]:

In [ ]:

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```

```